

Tech Layoffs Analysis Dashboard using Microsoft Excel

Project Overview

The **Tech Layoffs Analysis Dashboard** is an interactive Excel dashboard developed to analyze global technology layoffs from **2020 to 2025**. The project transforms raw layoff data into meaningful business insights using Microsoft Excel's data analysis and visualization features.

The objective of this project is to identify layoff trends across different years, companies, industries, countries, and months while providing an interactive dashboard for easy exploration and decision-making.

Objectives

- Analyze global technology layoffs between 2020 and 2025.
 - Identify the companies with the highest layoffs.
 - Compare layoffs across different countries and industries.
 - Study yearly and monthly layoff trends.
 - Build an interactive dashboard using Excel.
 - Demonstrate data cleaning, analysis, and visualization skills.
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Dataset Information

- **Source:** Kaggle
- **Dataset:** Tech Layoffs (2020–2025)
- **Rows:** 2,412
- **Columns:** 18

The dataset includes:

- Company
- Headquarters Location
- Country
- Continent
- Industry

- Company Stage
 - Employees Laid Off
 - Layoff Percentage
 - Date of Layoff
 - Company Size
 - Money Raised
 - Year
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Data Cleaning Process

Before creating the dashboard, the dataset was cleaned to improve data quality.

The following steps were performed:

- Converted the dataset into an Excel Table.
 - Checked and removed duplicate records.
 - Standardized date formatting.
 - Filled missing values in the **Stage** column with "**Unknown**".
 - Retained missing values in fields where information was unavailable instead of making assumptions.
 - Created helper columns such as Month and Year for trend analysis.
 - Verified data consistency before analysis.
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Dashboard Features

The dashboard contains interactive KPI cards and Pivot Charts, including:

KPI Cards

- Total Employees Laid Off
- Total Companies
- Total Countries
- Total Industries

Visualizations

- Annual Layoff Trend
- Top 10 Companies by Layoffs
- Top 10 Countries by Layoffs
- Industry-wise Layoff Analysis
- Monthly Layoff Trend

Interactive Filters

- Year
- Country
- Industry
- Month
- Continent

These slicers allow users to explore the dashboard dynamically.

Key Insights

- More than **746,000 employees** were laid off during the analyzed period.
 - The dataset contains **1,706 companies** across **49 countries** and **115 industries**.
 - Layoffs increased significantly during **2023**, indicating one of the largest workforce reductions in the technology sector.
 - The **United States** experienced the highest number of layoffs.
 - Large technology companies contributed significantly to overall layoffs.
 - Certain industries experienced considerably higher workforce reductions than others.
 - Monthly analysis reveals noticeable fluctuations, reflecting changing economic conditions.
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Tools & Skills Used

Software

- Microsoft Excel 2024

Excel Features

- Pivot Tables
 - Pivot Charts
 - Slicers
 - Excel Tables
 - Data Cleaning
 - Sorting & Filtering
 - UNIQUE() Function
 - COUNTA()
 - Basic Excel Formulas
 - Dashboard Design
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Business Value

This dashboard enables users to:

- Monitor workforce reduction trends.
 - Compare layoffs across companies, countries, and industries.
 - Identify periods with the highest layoffs.
 - Support business reporting through interactive visualizations.
 - Improve decision-making using data-driven insights.
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Conclusion

This project demonstrates the complete workflow of an Excel-based data analysis project—from data cleaning and preparation to interactive dashboard development. It highlights practical skills in data visualization, business reporting, and analytical thinking, making it a valuable addition to a data analyst portfolio.